

WT32-ETH01 DMX Lighting Node

User Manual & Integration Guide

Firmware Version 1.30.00

Project Documentation

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Quick Start

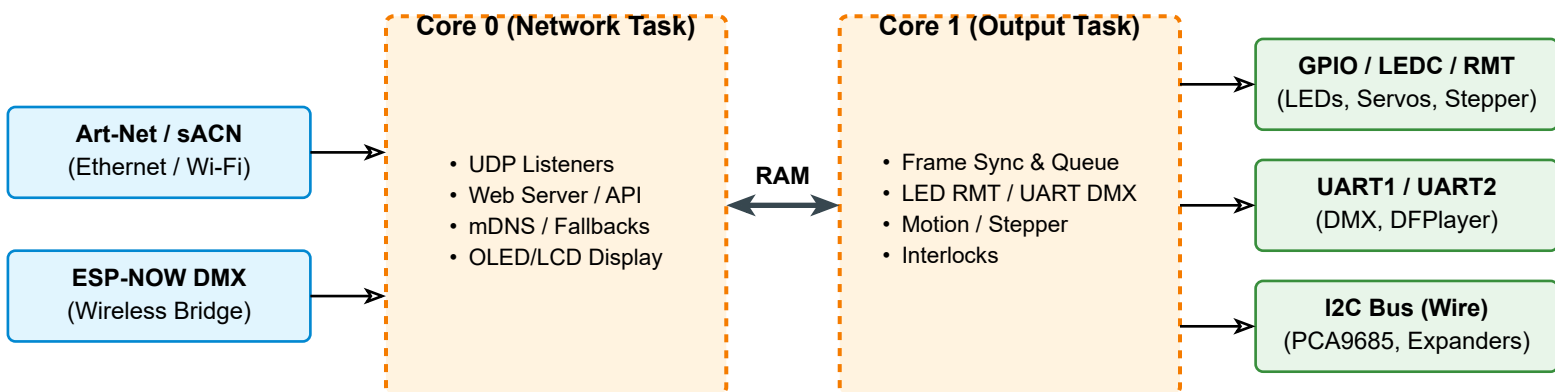
To begin using the WT32-ETH01 DMX Lighting Node, follow these steps:

1. **Power the Node:** Connect a stable 5V power supply to the WT32-ETH01 board. Ensure the supply can deliver sufficient current for the ESP32, Ethernet transceiver, and any connected outputs.
2. **Initial Setup AP:** If Ethernet is disconnected and Wi-Fi client mode is not yet configured, the board automatically starts a configuration Access Point named ESP32-ArtNet-Setup-XXXX.
3. **Access Configuration:** Connect your PC or mobile device to the setup AP and navigate to <http://192.168.4.1> in a web browser.
4. **Configure Settings:** Navigate through the Network, Outputs, and Protocol tabs to define your system parameters.
5. **Save and Apply:** Click **Save**. The board will validate the settings against hardware constraints and reboot.

Ethernet is strongly recommended for live Art-Net/sACN streaming. Wi-Fi client mode is useful for setup or light-duty/non-critical streaming.

System Architecture

The system architecture utilizes the ESP32's dual-core processor to separate high-priority output timing from network protocol processing. The diagram below illustrates this runtime task split and signal flow.



Network Modes & System Identity

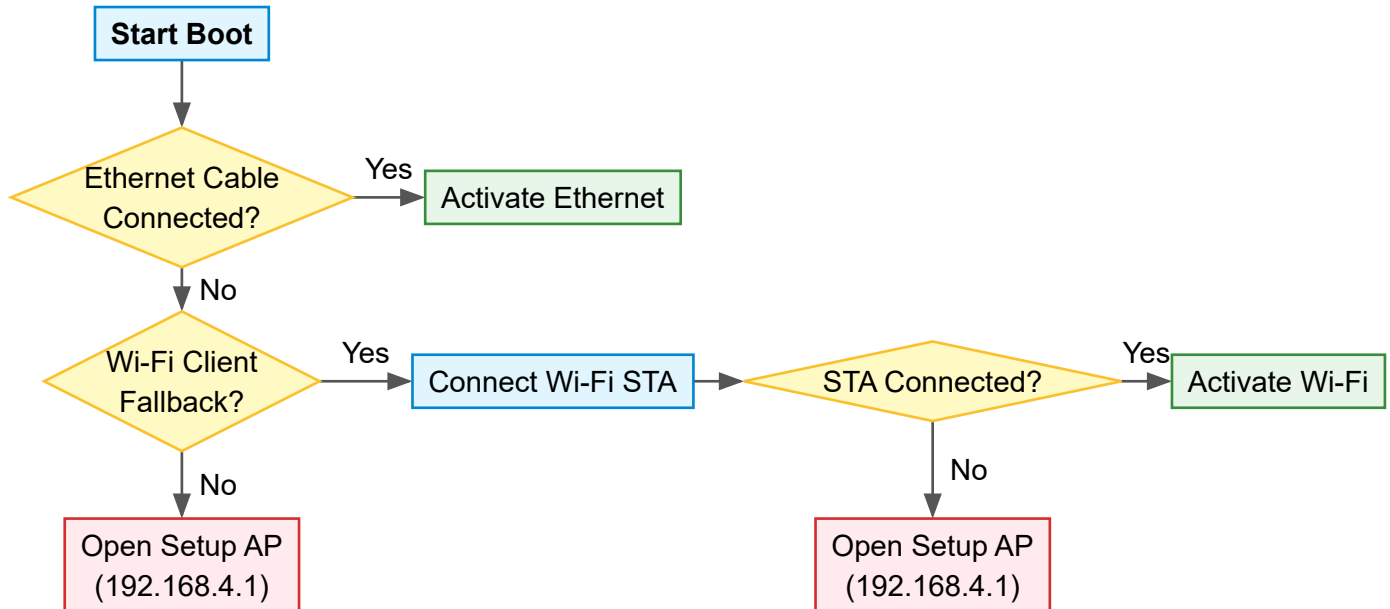
The WT32-ETH01 firmware supports multiple network modes and automated fallbacks to maintain network responsiveness under varying installation conditions.

Network Modes

- **Art-Net Ethernet (Wired):** The main operating mode. The device obtains an IP address via DHCP or static configuration and listens for Art-Net UDP packets on the default port 6454.
- **ESP-NOW Master:** Receives DMX inputs via Art-Net/sACN and encodes the stream into custom wireless packets sent to remote slave nodes.
- **ESP-NOW Slave:** Receives wireless DMX streams directly from an ESP-NOW Master and drives local output peripherals.

Network Fallback System

During boot in Art-Net Ethernet mode, the device executes a priority-based fallback logic to ensure communication remains accessible if cables are disconnected or signals are weak. The flowchart below outlines this sequence.



System Identity & MAC Suffix

To prevent network clashes when multiple lighting nodes are deployed on the same subnet, the device generates distinct identifiers based on its unique MAC address:

- **Setup AP SSID:** ESP32-ArtNet-Setup-XXXX (where XXXX is the last 4 hex characters of the MAC address).
- **mDNS Hostname:** http://[hostname]-[xxxx].local (where the suffix is lowercase).
- **ArtPollReply Names:** Short Name: CHAL Node-XXXX; Long Name incorporates the board type and suffix.

Boot Resiliency & Recovery Mode

If invalid configurations or networking issues cause the firmware to crash or boot loop, the device provides a hardware and software-based **Recovery Mode**:

1. **5-Power Cycle Trigger:** Powering the device on and off consecutively 5 times (rebooting within 15 seconds of boot) forces the board into Recovery Mode. A watchdog timer resets this count to 0 after 15 seconds of stable runtime.
2. **Physical Button Trigger:** Hold the physical **BOOT button (GPIO0)** low during power-up.

When Recovery Mode is activated:

- **Dual Network Access:** The node spins up both Ethernet and an open (no password) Wi-Fi AP named ESP32-ArtNet-Recovery-XXXX.
- **mDNS Resolution:** The recovery interface resolves at http://[hostname]-recovery.local (with no MAC suffix, simplifying access).
- **Low Power State:** All physical output signals and DMX listening loops are suspended to avoid brownout conditions.

Status LED Blink Codes

The Status LED (default GPIO 5) communicates the system state through distinct repeating flash sequences:

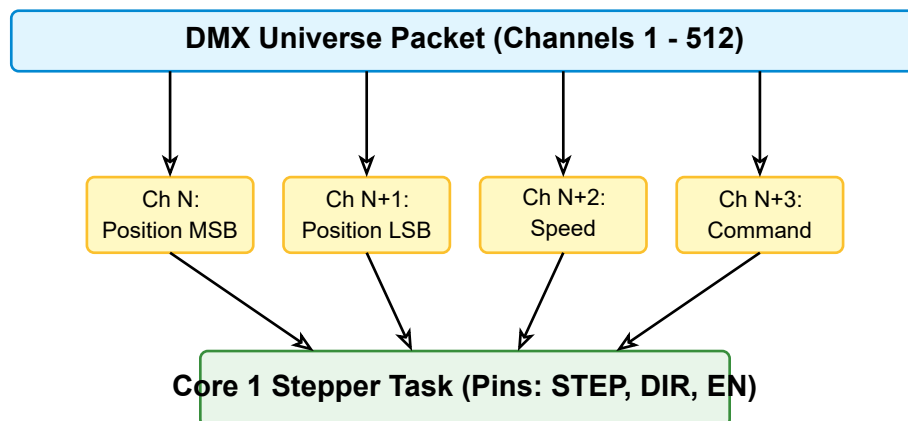
Blinks	Device State
1 Blink	Disconnected or network interface idle
2 Blinks	SoftAP configuration portal active
3 Blinks	Ethernet link established and active
4 Blinks	Wi-Fi STA connection established and active
5 Fast Blinks	ESP-NOW Slave receiving recent valid packets
6 Fast Blinks	Manual Output Test mode active (overriding DMX stream)

Outputs & Capacity Planning

The node allocates outputs dynamically according to a hardware budget scoring algorithm. Output channels map DMX data to physical pins using multiple sources.

Output Concepts

Each output channel is defined by a start universe, start DMX address, type (0–18), source (GPIO, PCA9685, digital expander, I2C DAC), and parameters.



Resource & Compute Scoring System

To guarantee dual-core stability, the firmware checks **three independent budgets** before accepting a configuration. If any one is exceeded, saving is blocked:

Budget	Limit	What It Counts
HardwareResource	LEDC ≤ 16,	Finite ESP32 peripherals (PWM, serial, etc.) — expander/PCA channels bypass this count

	$\text{RMT} \leq 8,$ $\text{UART} \leq 2,$ $\text{DAC} \leq 2,$ $\text{Timer} \leq 4$	
CpuBudget	$\leq \left(1,000, \frac{000}{\text{fps}}\right)$ 1,500 μs per frame	Total output service time including DMX transmit, pixel mapping, I2C writes, and 500 μs base overhead
RamBudget	$\leq 64 \text{ KB}$	Static buffer estimates per channel plus runtime driver objects; leaves 150 KB free for system/network

Per-Type Resource Footprint

Each output type has estimated CPU service time and RAM usage. Refer to the full tables in docs/resource_calculator.md for exact $\mu\text{s}/\text{frame}$ and byte estimates by type, source, and routing mode.

Output Type Reference

ID	Type Name	Bytes	Compatible Sources	Primary Resource
0	AC Dimmer	1–2	GPIO Only	GPIO 1 + ZC Timer
1	DMX Serial Output	512	GPIO Only (UART/RMT)	UART (fallback RMT)
2	Relay (On/Off)	1	GPIO, PCA, Expander	GPIO / PCA / Exp 1
3	RGB/RGBW LED Strip	Var	GPIO Only (RMT)	RMT 1 ch
4	Single Color LED	1–2	GPIO, PCA	LEDC 1 / PCA 1
5	Analog RGB/RGBW	3–4	GPIO, PCA (Hybrid)	LEDC 3-4 / PCA 3-4
6	DC Motor (H-Bridge)	1–2	GPIO, PCA (Hybrid)	LEDC 1-3 / PCA 1-3
7	Stepper Motor	3–5	STEP:GPIO; DIR:Any	FastAccelStepper
8	RC Servo	1	GPIO, PCA	LEDC 1 (50Hz) / PCA 1
9	Passive Buzzer	2	GPIO Only	LEDC 1
10	DFPlayer MP3 Module	3	GPIO Only (UART)	UART 1
11	7-Segment (TM1637)	2–4	GPIO Only	GPIO 2

12	7-Segment DD 7-Pin	1–2	GPIO, PCA, Ex- pander	Seg-level routing
13	7-Segment DD 8-Pin	1–2	GPIO, PCA, Ex- pander	Seg-level routing
14	DAC (Analog Out)	1	MCP4725 (src 5), DAC7571 (src 6), DAC7573 (src 7)	I2C Bus
15	PWM DAC (RC Filter)	1	GPIO, PCA	LEDC 1 / PCA 1
16	Function Generator	5	GPIO Only	LEDC 1 + timer
17	Solenoid Trigger	1	GPIO, PCA, Ex- pander	GPIO / PCA / Exp 1
18	Smoke Shooter	1	GPIO, PCA, Ex- pander	GPIO / PCA / Exp 2

Pin Interlocks

To maintain hardware safety, configuration rules enforce the following constraints:

- **No GPIO Duplication:** An ESP32 GPIO pin cannot be used by more than one output channel or system setting simultaneously.
- **System Reservation:** GPIO pins configured as Status LED, Zero-Crossing Input, or I2C SDA/SCL are blocked from output selection.
- **PCA9685 Shared Frequency:** All channels on a single PCA9685 chip share one output frequency. Selecting an RC Servo (Type 8) on a PCA9685 chip automatically forces all channels on that chip to 50 Hz.
- **Stepper STEP Pin:** The STEP signal for Type 7 (Stepper Motor) must use a hardware-capable ESP32 GPIO pin due to high frequency toggle requirements. Enable, DIR, and Limit pins can route to digital expanders.
- **Motor EN Pin:** DC Motors operating in Mode 2 (IN1+IN2+EN) require speed modulation on the EN pin. Therefore, the EN pin must route to an ESP32 GPIO or a PCA9685, never a digital expander.

PWM DAC Duty Calibration

For Type 15 (PWM DAC) outputs, you can map the digital 0–255 DMX range to specific hardware duty cycles. This is configured via:

- **Mode:** Custom, 0–10V output converter, or 4–20mA transmitter.
- **Min Duty:** Duty cycle percentage multiplied by 100 (e.g. 500 = 5.0%) mapped to DMX 0.
- **Max Duty:** Duty cycle percentage multiplied by 100 (e.g. 9500 = 95.0%) mapped to DMX 255.

Linear interpolation mapping runs inside Core 1's update loop to set the LEDC duty register correctly.

ESP-NOW Wireless DMX Bridge

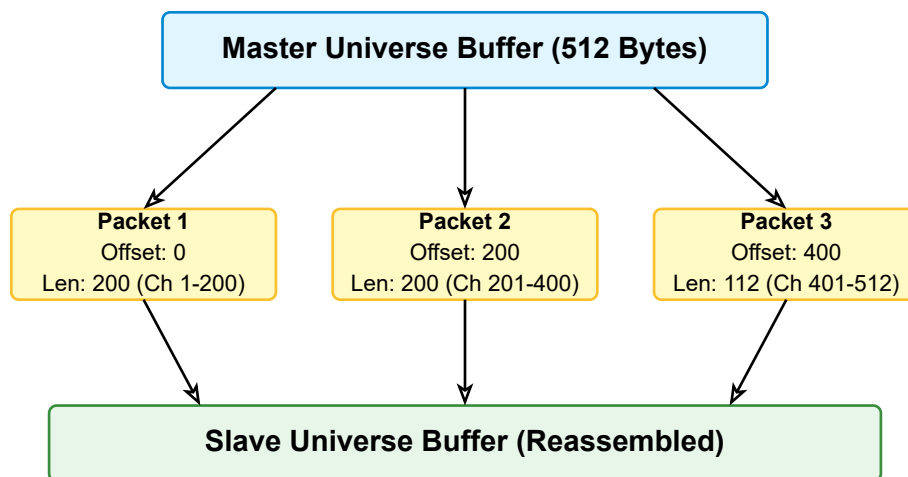
The ESP-NOW Wireless Bridge allows wireless forwarding of DMX universes from a network-connected Master board to one or more remote Slave boards.

Recommended Setup Flow

1. **Configure Slave:** Set the remote board to ESP-NOW Slave mode. Access its Web UI, navigate to the ESP-NOW tab, and copy its physical Wi-Fi MAC Address.
2. **Configure Master:** Set the central network-connected board to ESP-NOW Master mode. Under the ESP-NOW Peer List, click **Add Peer**.
3. **Define Routing Range:** Paste the MAC Address and input the start/end universe and channel ranges mapping to this slave.
4. **Save and Sync:** Save settings. The Master starts encoding and transmitting matching DMX packets.

Universe Chunking & Custom Protocol

ESP-NOW frames have a physical payload limit of 250 bytes. To transmit a full 512-channel DMX universe, the Master splits the universe data into chunks using a configurable chunk size (16–230 bytes, default 200). Each chunk is prefixed with a 12-byte custom header followed by DMX data bytes.



Payload Layout

The custom firmware packet header consists of:

1. Header (4 bytes): ASCII "DMX\0"
2. Universe (2 bytes): 16-bit unsigned integer (0..32767)
3. Offset (2 bytes): 16-bit unsigned integer (0..511)
4. Total Length (2 bytes): 16-bit unsigned integer (1..512)
5. Length (2 bytes): 16-bit unsigned integer (1..230) — number of DMX data bytes in this chunk

Offset-Based Chunking Protocol

Each chunk is a **self-contained packet** — the Slave does not reassemble the full universe before driving outputs. Instead, every incoming chunk is applied immediately by computing the overlap between the chunk's content range and each channel's DMX address range.

Master: Partitioning

For each peer's configured routing range, the Master walks the DMX universe buffer in steps of chunkSize:

Chunk	Offset	Length	DMX Channels
1	0	200	1–200
2	200	200	201–400
3	400	112	401–512

The loop produces packets where offset advances by chunkSize each iteration and length is clamped so the last chunk does not exceed the universe boundary.

If a peer's address range is narrower than the full universe (e.g., Ch 10–30), the Master computes `firstOffset = 9` and `lastOffsetExclusive = 30`. Only overlapping bytes are sent — no RF airtime is wasted on channels outside the peer's range.

Slave: Overlap Mapping

Every received chunk is pushed to a FreeRTOS queue and processed on Core 1. For each active output channel, the Slave finds the intersection of:

- The chunk's data window: `["offset" , "offset" + "length" ]`
- The channel's DMX source window: `[startAddress - 1, startAddress - 1 + bufferSize)`

If the windows overlap, the intersecting bytes are copied into the channel's DMX value buffer. All other channels are left untouched by this packet. Because each chunk is processed independently, packets may arrive out of order or be lost — the latest received value for any given DMX slot always wins.

Example: A Slave configured with two outputs — Relay at Ch 10 and Single LED at Ch 250 — receives three chunks from a peer with range Ch 1–512. Only the relay channel receives bytes from Chunk 1; only the LED channel receives bytes from Chunk 2. Chunk 3 is discarded by both. No per-channel airtime is wasted.

Sequence & Loss Handling

The protocol has no retransmission or sequence numbering. If a chunk is lost in the RF medium, the affected DMX slots simply hold their last received value (Hold Last State per ADR005). On the next frame cycle, the Master re-sends the same universe — lost slots recover within one frame period.

Network Stability & Planning

To maximize RF efficiency and reliability, design your layouts around these operational constraints:

- **Narrow Peer Ranges:** Map only the specific channels a slave requires. If a slave only controls a single relay at channel 10, define its peer range on the Master as `u0:CH10` to `u0:CH10`. This prevents transmitting unused DMX bytes.
- **Broadcast Fallback:** If no peers are defined on the Master, it falls back to broadcasting packets to `FF:FF:FF:FF:FF:FF`. Use this only for testing, as broadcast packets consume high airtime and increase collision rates in crowded RF environments.

- **Queue Deferral:** Under the hood, the ESP-NOW callback on Core 0 pushes incoming frames to a FreeRTOS queue. Core 1's output task handles the de-queueing and physical routing. This prevents blocking Core 0's Wi-Fi radio stack.

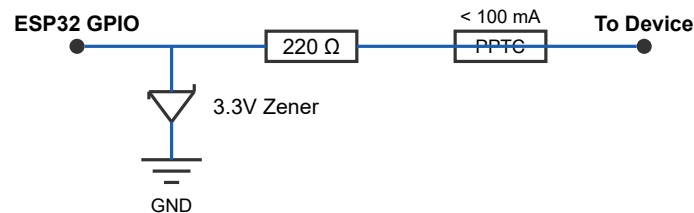
Hardware Protection & Verification

When the board drives various loads for lighting and effect control in the field, current surges or noise feedback can cause brownout resets or port damage. This chapter documents the recommended protection circuits and device verification guidelines.

Output Port ESD & Overcurrent Protection

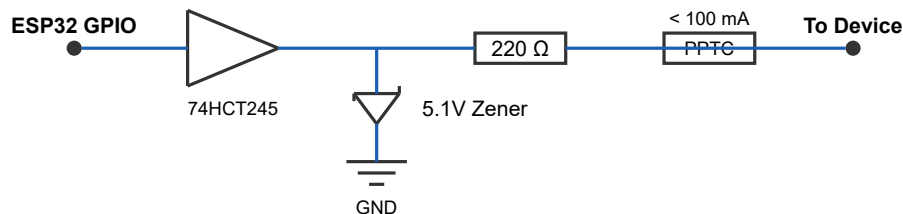
Direct GPIO Connection (3.3V)

Use a 3.3V Zener diode to prevent the ESP32 signal pin voltage from exceeding the 3.3V rating when driving a load directly.



Buffered Connection (5V)

When using a 5V buffer IC (e.g., 74HCT245), use a 5.1V Zener diode on the buffer output side.



Component Details

Component	Value	Purpose
Zener Diode (ESD Clamp)	3.3V (GPIO) / 5.1V (Buffered)	Clamp excess voltage; protect against ESD or reverse voltage
Series Resistor	47 Ω to 220 Ω (see below)	Limit maximum current on signal line
PPTC Resettable Fuse	30 mA to 100 mA (see below)	Resettable overcurrent protection on output line

Component Selection for 12V/24V Fault Tolerance

If the board operates alongside 12VDC lighting or 24VDC motor power supplies, accidental short-circuits can inject these higher voltages back into the signal lines. To prevent damage, the series resistor (R) and PPTC fuse must be selected so that the fault current (I_{fault}) exceeds the PPTC trip current (I_{trip}) to trigger a shutdown, while limiting the transient power dissipation on the Zener diode.

The fault current is calculated as:

$$I_{\text{fault}} = \frac{V_{\text{fault}} - V_{\text{Zener}}}{R + R_{\text{PPTC}}}$$

Recommended Configurations for 12V and 24V Coexistence

1. 50 mA Hold / 100 mA Trip PPTC (Recommended for general use & pixel strips):

- **Resistor Value (R):** 47 Ω to 82 Ω (Use a 0.5W or 1W pulse-withstanding resistor, e.g. size 1206 or 1210).
- **At 12V Fault ($V_{\text{fault}} = 12\text{V}$):** $I_{\text{fault}} \approx 82\text{mA} - 146\text{mA}$, which successfully trips the 100 mA fuse.
- **At 24V Fault ($V_{\text{fault}} = 24\text{V}$):** $I_{\text{fault}} \approx 220\text{mA} - 350\text{mA}$, causing an instantaneous trip.
- **Note:** Low series resistance maintains high rise/fall times for WS2812B pixels (800 kHz) and DMX512 (250 kbps).

2. 30 mA Hold / 60 mA Trip PPTC (Recommended for slow signal lines & relays):

- **Resistor Value (R):** 100 Ω to 120 Ω .
- **At 12V Fault:** $I_{\text{fault}} \approx 60\text{mA} - 85\text{mA}$, which trips the 60 mA fuse.
- **At 24V Fault:** $I_{\text{fault}} \approx 150\text{mA} - 200\text{mA}$, causing a rapid trip.

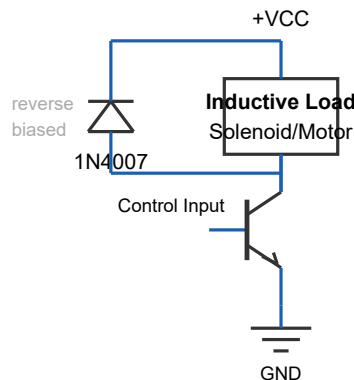
3. 20 mA Hold / 40 mA Trip PPTC (For high-impedance inputs only):

- **Resistor Value (R):** 150 Ω to 220 Ω .
- **Note:** Not recommended for high-speed pixel strips due to signal rounding from higher RC time constants.

Inductive Load Protection (Flyback Diode)

Inductive loads (solenoid valves, relay coils, DC motors) generate a high reverse voltage spike (flyback voltage) when suddenly switched off, which can disrupt the radio system or damage the driving transistor.

Mitigation: Always connect a fast or general-purpose diode (e.g., 1N4007) in reverse-biased parallel across the inductive load to safely discharge the reverse voltage.

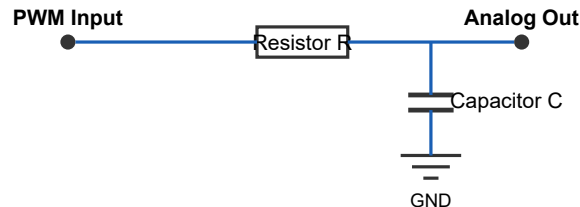


PWM DAC & Function Generator Low-Pass Filter

To convert high-frequency PWM outputs into smooth analog DC voltages (for PWM DAC mode) or analog waveforms (for Function Generator mode), an external Resistor-Capacitor (RC) Low-Pass Filter (LPF) is required.

Passive 1st-Order RC Filter

The simplest way to filter a PWM signal is using a passive RC filter.



The cutoff frequency (f_{cutoff}) is determined by the formula:

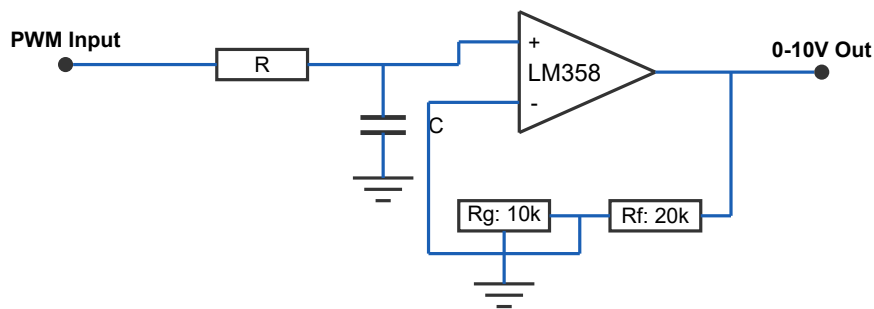
$$f_{\text{cutoff}} = \frac{1}{2\pi RC}$$

Component Selection by Mode:

- **PWM DAC Mode (Mode 15 - DC Voltage):** Recommended cutoff is 1.6Hz – 16Hz to minimize voltage ripple. Use $R = 10\text{k}\Omega$ and $C = 1\mu\text{F}$ ($f_{\text{cutoff}} \approx 16\text{Hz}$) or $C = 10\mu\text{F}$ ($f_{\text{cutoff}} \approx 1.6\text{Hz}$).
- **Function Generator Mode (Mode 16 - Waveforms up to 5 kHz):** Recommended cutoff is 7kHz – 10kHz to pass the 5 kHz signal while filtering the 50 kHz carrier. Use $R = 2.2\text{k}\Omega$ and $C = 10\text{nF}$ ($f_{\text{cutoff}} \approx 7.2\text{kHz}$).

Active 0-10V Buffer/Amplifier (LM358)

Passive RC filters have high output impedance and cannot drive significant loads directly. To output an industrial 0-10V signal from the ESP32's 3.3V PWM output, an active non-inverting amplifier using an op-amp (e.g., LM358) is recommended.



Gain Calculation:

$$\text{Gain} = 1 + \frac{R_f}{R_g} = 1 + 20\text{k}\frac{\Omega}{10\text{k}\Omega} = 3.0$$

(Input $3.3\text{V} \times 3.0 = 9.9\text{V}$ max output). For exactly 10.0V, replace R_f with a $20\text{k}\Omega$ multi-turn potentiometer.

- **Op-Amp VCC:** Must be powered by +12V to +24V for adequate headroom to reach +10V.

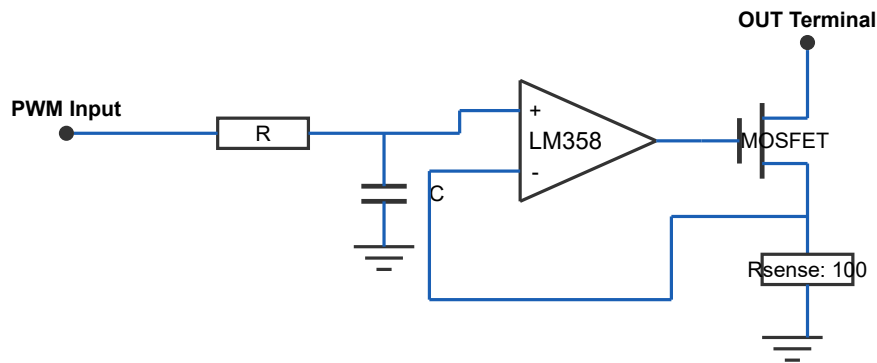
Active 0-5V and 0-24V Buffer/Amplifier (LM358)

To output other standard industrial voltage ranges like 0-5V or 0-24V from the ESP32's 3.3V PWM output, the same non-inverting amplifier topology is used by adjusting the feedback resistor values (R_f and R_g) and the Op-Amp supply voltage (VCC):

- **0-5V Analog Output:** Configure $R_g = 10\text{k}\Omega$, $R_f = 5.1\text{k}\Omega$ (Gain = 1.51, output up to $\approx 4.98\text{V}$). Power the Op-Amp with at least +7V (+12V is recommended).
- **0-24V Analog Output:** Configure $R_g = 10\text{k}\Omega$, $R_f = 62\text{k}\Omega$ (Gain = 7.2, output up to $\approx 23.76\text{V}$). For exactly 24V, replace R_f with a $50\text{k}\Omega$ trimmer in series with a $47\text{k}\Omega$ resistor. Power the Op-Amp with at least +26V.

Active 4-20mA Current Loop Transmitter

To output a standard 4-20mA current-loop signal, a voltage-controlled current sink using an Op-Amp and a power N-MOSFET (or NPN Darlington transistor like TIP122) in the feedback loop is recommended.



- **Operating Principle:** The loop load is connected between +24V loop power and the OUT Terminal. The output current is governed by the relation $I_{\text{out}} = \frac{V_{\text{in}}}{R_{\text{sense}}}$. With $R_{\text{sense}} = 100\Omega$:
 - $V_{\text{in}} = 0.4\text{V} \Rightarrow I_{\text{out}} = 4\text{mA}$
 - $V_{\text{in}} = 2.0\text{V} \Rightarrow I_{\text{out}} = 20\text{mA}$
- **Software Calibration:** In the Web UI settings for the PWM DAC channel, set `pwm_dac_min` to 1212 (representing 12.12% duty cycle, $\approx 0.4\text{V}$) and `pwm_dac_max` to 6060 (representing 60.60% duty cycle, $\approx 2.0\text{V}$) to calibrate the 4-20mA signal window.

Device Reference

All external peripheral devices must have reference datasheets for driver code verification.

\#	Device	Bus	Purpose / Verify
1	PCA9685	I2C	16-channel PWM controller; register map and frequency limits
2	MCP23017 / TCA9555	I2C	16-bit I/O expander; IODIR and output bitmap writes
3	PCF8574	I2C	8-bit I/O expander; quasi-bidirectional I/O and active-low behavior
4	MCP4725 / DAC7571 / DAC7573	I2C	12-bit DACs; byte-level commands and I2C address config

5	TM1637	GPIO	7-segment LED driver; CLK/DIO timing constraints
6	DFPlayer Mini / YX5200	UART	Audio player; 9600 bps command packet format
7	LAN8720A	RMII	Ethernet PHY; bootstrap pins and 50 MHz RMII clock GPIO impact
8	SSD1306 / SH1106	I2C	OLED display; I2C address and initialization sequence

I2C Bus Speed Compatibility

Device	Max I2C Speed
PCA9685	1 MHz (Fast-mode Plus)
MCP23017	400 kHz (Fast-mode)
PCF8574	100 kHz (Standard-mode)

Runtime Rule: The overall bus speed must not exceed the maximum supported by the slowest device on the bus, to avoid bus lockup or data corruption.

Logic Level Tolerances

- ESP32 operates at 3.3V logic levels; many relay modules and output drivers operate at 5V.
- Verify from the datasheet whether the 5V device accepts a 3.3V input signal as logic HIGH ($V_{IH} \leq 2.0 \text{ V}$), or whether a logic level shifter is required.

Timing & Start-up Delays

- Check the power-on reset / boot time for each IC.
- Some expander chips require a settling delay before accepting the first command, to prevent startup configuration errors.

Troubleshooting & Integration Guide

This section details standard debug procedures for hardware integrations, power issues, and software setup.

Common Issues

Device Configuration Portal is Unreachable

- **Symptoms:** Unable to access the Web UI at 192.168.4.1 or the device's Ethernet IP.
- **Solutions:**
 1. Ensure Ethernet activity lights are blinking on the WT32-ETH01 RJ45 jack.
 2. Verify that your PC/phone is connected to the SoftAP SSID ESP32-ArtNet-Setup-XXXX.
 3. Trigger Recovery Mode by holding down the **BOOT (GPIO0)** pin during boot, then access the recovery interface.

Only the First Part of an LED Strip Lights Up

- **Symptoms:** Pixels at the start of the strip function correctly, but remaining sections stay dark or freeze.
- **Solutions:**
 1. Check that the **LED Count** field in the channel's output configuration matches the number of physically wired pixels.
 2. Verify that your DMX controller/media server is sending all required universes. For example, a strip with 300 RGB LEDs requires 2 consecutive universes (e.g. Universe 0 and Universe 1).

Brownout Detector Resets (Boot Loops)

- **Symptoms:** The board boot loops immediately after powering up, with console prints showing Brownout detector was triggered.
- **Solutions:**
 1. **Separate Power Buses:** Do not draw power for high-current loads (LED strips, motors, relays, or solenoids) directly from the WT32-ETH01's 3.3V or 5V regulator pins.
 2. **Ground Ties:** Run a dedicated external power supply for high-current loads, and ensure its Ground (GND) terminal is connected to the WT32-ETH01 Ground pin.
 3. **Capacitive Filtering:** Add a 100 μ F to 1000 μ F capacitor across the power lines near high-load devices (like servo or stepper drivers) to buffer transient current spikes.

Integration Warnings

- **GPIO 12 Bootstrapping Warning:** GPIO 12 (MTDI) is a bootstrapping pin that controls the internal flash voltage regulator state during boot. If an external peripheral pulls this pin High or Low incorrectly during startup, the ESP32 will fail to read its flash chip and fail to boot. Ensure any circuit connected to GPIO 12 is in a high-impedance (high-Z) state during power-up.
- **GPIO 16 Reservation:** GPIO 16 must not be selected for user outputs under any configuration. It controls the LAN8720 Ethernet transceiver power state. Overriding it will break network connectivity.